

SDLC & Agile Methodology

Self-Study Guide for Trainees

Week 1: SDLC Fundamentals

- Understand what SDLC is and why organizations use it.
- Learn the seven SDLC phases.
- Identify deliverables and stakeholders for each phase.
- Activity: Map the SDLC phases for an online shopping application.
- Self-check: Explain each SDLC phase in your own words.

Reflection Questions

1. What did I learn this week?
2. Which concepts need revision?
3. Can I explain this topic without referring to notes?
4. Can I solve scenario-based questions?

Week 2: SDLC Models

- Study Waterfall, V-Model, Incremental, Iterative, Spiral and Agile.
- Compare advantages, disadvantages and suitable projects.
- Activity: Choose the best model for Banking, Healthcare and Start-up projects.
- Quiz: Why is Spiral useful for high-risk projects?

Reflection Questions

1. What did I learn this week?
2. Which concepts need revision?
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Week 3: Agile & Scrum

- Read the Agile Manifesto and the 12 Principles.
- Understand Scrum roles, events and artifacts.
- Create five user stories for a food delivery application.
- Activity: Prioritize a sample product backlog.

Reflection Questions

1. What did I learn this week?
2. Which concepts need revision?
3. Can I explain this topic without referring to notes?
4. Can I solve scenario-based questions?

Week 4: DevOps & CI/CD

- Learn the basics of Continuous Integration and Continuous Delivery.
- Understand the role of Git, Jenkins, Docker and Kubernetes.
- Draw a simple CI/CD pipeline.
- Mini Project: Design an Agile SDLC for an online banking system.

Reflection Questions

1. What did I learn this week?
2. Which concepts need revision?
3. Can I explain this topic without referring to notes?
4. Can I solve scenario-based questions?

Final Revision Checklist

- Explain all SDLC phases.
- Compare all SDLC models.
- Describe Agile values and principles.
- Explain Scrum roles, events and artifacts.
- Write user stories and acceptance criteria.
- Explain CI/CD and DevOps.
- Answer scenario-based interview questions confidently.